

# THE DEVELOPERS ARENA

The world where only developers exist



A

## ***Project Report***

*submitted in partial fulfilment*

*for the Internship of*

*B.TECH.*

*In*

*Department of*

## **COMPUTER SCIENCE AND ENGINEERING**

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Position: Data Science Intern

Internship Duration: 2 Months

Submitted To: *THE DEVELOPERS ARENA*

Submission Date: 07/07/2026

“Name of the Institute” –

Institute of Engineering & Technology, Udaipur

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# Task 4: Data Visualization and First complete Project

## Project Description:

### Project Title: E- Commerce Sales Analysis

-Create a full analysis on the topic E commerce sales analysis with at least two different types of charts and written insights.

## Project Overview

E-commerce businesses generate a large amount of transactional data every day. Analyzing this data helps organizations understand customer purchasing behavior, identify high-performing products, evaluate regional sales, and improve business decisions.

This project performs a complete analysis of an e-commerce sales dataset using Python. The project includes data preprocessing, exploratory data analysis, visualization, KPI calculation, and business insight generation.

## Project Objectives

The objective is to help business understand their sales performance, identify the top-selling products, analyze regional sales, and support data driven decision making.

The main objectives of this project are:

- Analyze overall sales performance.
- Perform data cleaning and processing.
- Calculate Key Performance Indicators (KPIs).
- Identify top- selling product.
- Analyze regional sales performance.
- Create meaningful visualizations.
- Generate written insights.

# Problem Statement

Businesses often collect large amounts of sales data but may struggle to convert it into actionable insights. This project aims to analyze an e-commerce sales dataset to understand product performance, regional sales trends, and customer purchasing patterns using data visualization and statistical analysis.

# Dataset Description

The dataset contains e-commerce transaction records.

| Column      | Description                |
|-------------|----------------------------|
| Date        | Date of Sale               |
| Product     | Product Name               |
| Quantity    | Number of Units Sold       |
| Price       | Price per Unit             |
| Customer_ID | Unique Customer Identifier |
| Region      | Sales Region               |
| Total_Sales | Total Revenue Generated    |

## Dataset Summary

- Total Records: 100
- Total Columns: 7
- File Format: CSV

## Technologies Used:

- Pandas
- Vs code
- Terminal
- Python
- Matplotlib
- Jupyter notebook

## Setup Instructions

### Step 1

Install Python 3.14.5

### Step 2

Clone or download the project.

### Step 3

Install required libraries.

Bash - `pip install -r requirements.txt`

### Step 4

Run the project:

- Python main.py
- Or
- Open the notebook – analysis.ipynb

# Project Structure on Github:

```
E-Commerce_Sales_Analysis_Project/
├── data/
│   └── sales_data.csv
├── report/
│   └── E-Commerce Sales Analysis Project.pdf
├── visualizations/
│   ├── monthly_sales_Line chart.png
│   ├── price_vs_sales_Scatter plot.png
│   ├── product_sales_Bar chart.png
│   ├── quantity_distribution_Histogram.png
│   └── region_sales_Pie chart.png
├── analysis.ipynb
├── main.py
├── README.md
├── requirements.txt
├── screenshots/
│   ├── code.png
│   └── output.png
```

## Code Structure Explanation

### 1. Importing the Library

```
import pandas as pd
import matplotlib.pyplot as plt
```

#### Pandas

Pandas is a Python library used for loading, cleaning, manipulating, and analyzing structured data efficiently using Data Frames.

## **Matplotlib**

Matplotlib is a Python visualization library used to create charts and graphs that help represent data in a clear and meaningful way.

### **2. Loading the Dataset**

The `read_csv()` function loads the sales dataset into a Pandas Data Frame for analysis. Display dataset information.

### **3. Checking Missing Values**

Detects null values in the dataset before performing the data analysis.

### **4. Data Cleaning**

Removes duplicate records and validates the dataset.

### **5. KPIs Analysis**

KPI Calculates:

- Total sales
- Total orders
- Total quantity
- Average sales
- Best-selling product

### **6. Product Analysis**

Analyzes total sales for each product to identify the highest-selling products and compare their performance.

### **7. Region Analysis**

Examines sales across different regions to identify the highest revenue-generating region and compare regional performance.

## **8. Monthly Sales**

Analyzes monthly sales trends to identify changes in revenue over time and seasonal patterns.

## **9. Visualizations**

Creates graphs using Matplotlib.

## **10. Business Insights**

Prints meaningful insights generated from the analysis.

# **Exploratory Data Analysis**

The Following analysis were performed:

### **Total Sales Analysis**

- Calculated the total revenue generated from all sales transactions.

### **Product Analysis**

- Compared sales generated by different product to identify top-selling products.

### **Regional Analysis**

- Compared total sales across different regions

### **Quantity Analysis**

- Analyzed customer purchasing quantities.

### **Price Analysis**

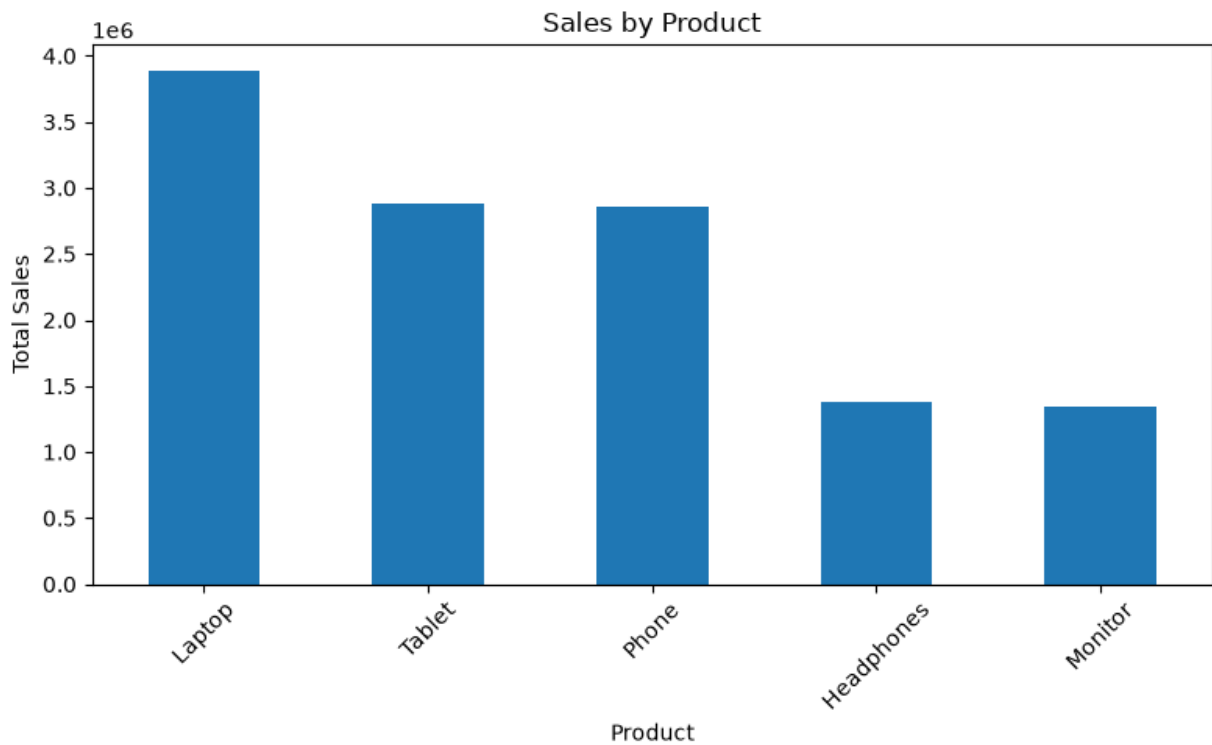
- Studied the relationship between price and total sales.



# Data Visualizations

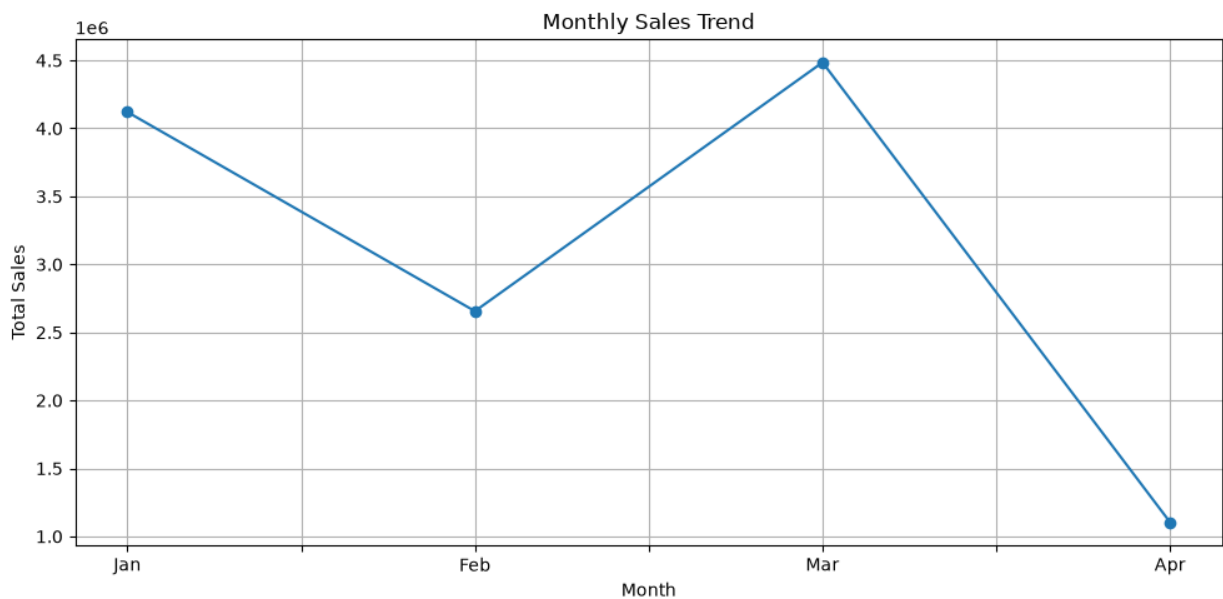
## a) Chart 1 : Product Sales (Bar chart)

- Compare total sales for each product.



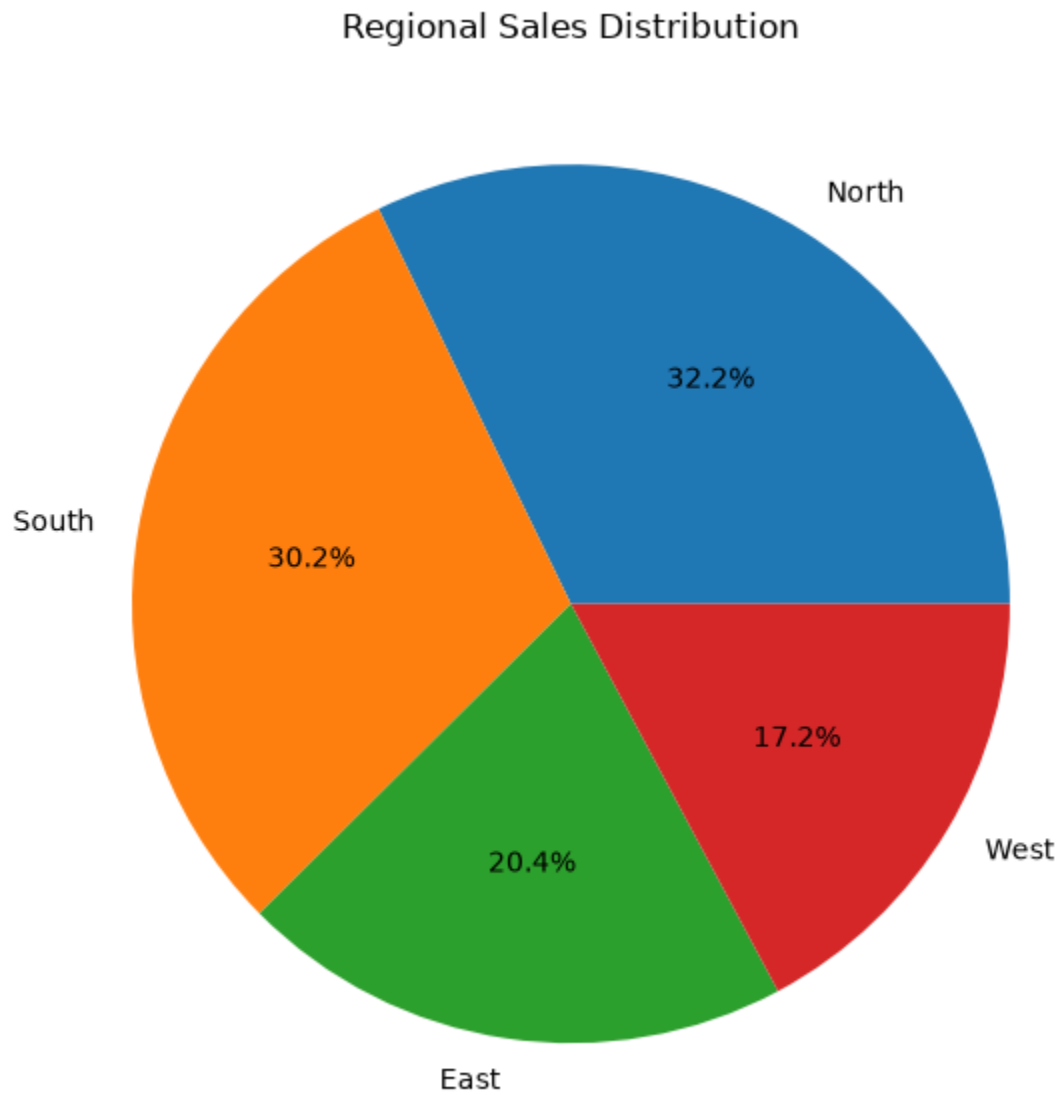
## b) Chart 2: Monthly Sales Trend (Line chart)

- Display sales trends over time.



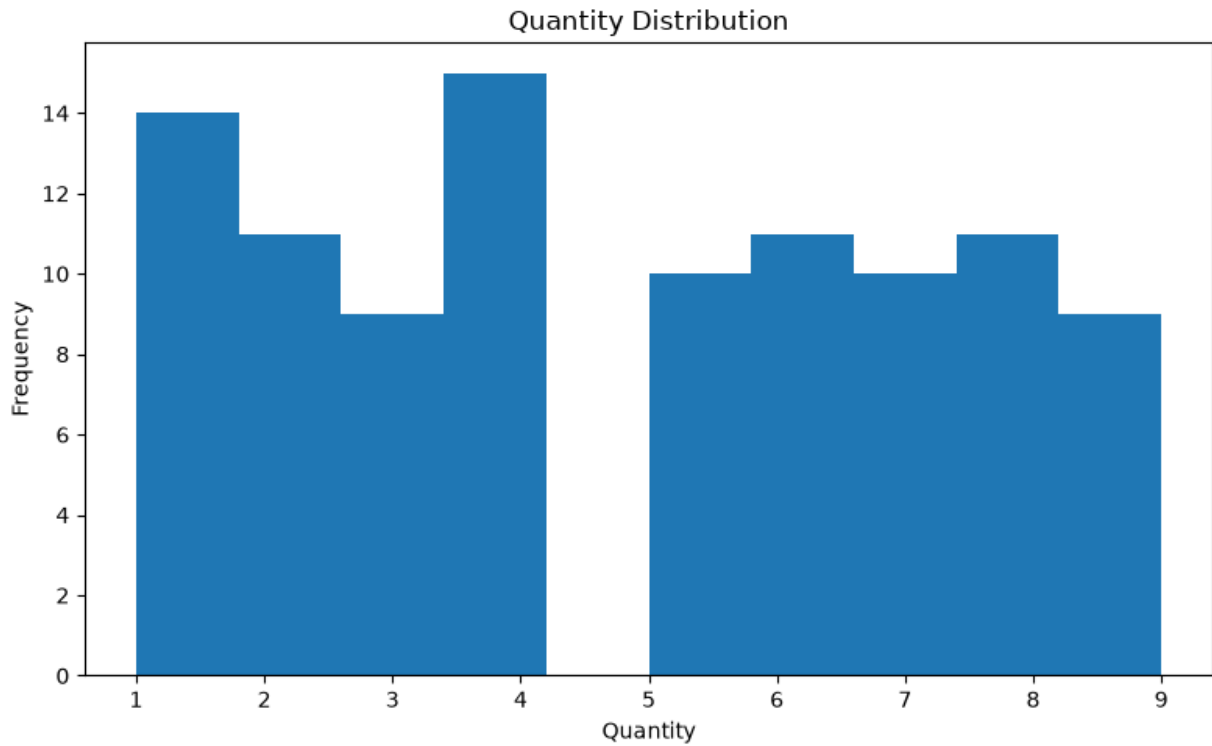
**c) Chart 3: Region Sales (Pie chart)**

- Show each region's contribution to total sales.



**d) Chart 4: Quantity Distribution (Histogram)**

- Display the frequency of purchase quantities.



**e) Chart 5: Product vs Total Sales (Scatter Plot)**

- Visualize the relationship between product price and revenue.



## **Business Insights**

- Identified the highest-selling product based on total revenue.
- Compared sales performance across different regions.
- Analyzed customer purchasing patterns through quantity distribution.
- Evaluated the relationship between product price and total sales.
- Generated actionable insights to support business decision-making.

## **Testing Evidence**

The project was tested successfully by:

- Running the Python script without execution errors.
- Verifying dataset loading.
- Validating KPIs calculations.
- Confirming that all visualizations were generated correctly.
- Checking that charts images were saved successfully.
- Ensuring outputs matched the expected results.

# Screenshots of working applications

```
main.py > ...
1  import pandas as pd
2  import matplotlib.pyplot as plt
3
4
5  # LOAD DATASET
6
7  df = pd.read_csv("data/sales_data.csv")
8
9  print("="*50)
10 print("First 5 Rows")
11 print("="*50)
12 print(df.head())
13
14 print("\nDataset Information")
15 print(df.info())
16
17
18 #Checking missing values
19
20 print("\nMissing Values")
21 print(df.isnull().sum())
22
23
24 # DATA CLEANING
25
26 # Remove duplicate rows
27 df.drop_duplicates(inplace=True)
28
29 # Convert Date column to datetime
30 df['Date'] = pd.to_datetime(df['Date'])
31
32 # Create Month column
33 df['Month'] = df['Date'].dt.strftime('%b')
34
35 print("\nSummary Statistics")
36 print(df.describe())
37
```

```

38
39 # KPI ANALYSIS
40
41 total_sales = df['Total_Sales'].sum()
42 total_orders = len(df)
43 total_quantity = df['Quantity'].sum()
44 average_sales = df['Total_Sales'].mean()
45
46 #Best Selling Product
47 best_product = (
48     df.groupby("Product")["Quantity"]
49     .sum()
50     .idxmax()
51 )
52
53 print("\n===== KPIs =====")
54 print(f"Total Sales: ₹{total_sales:,.2f}")
55 print(f"Total Orders: {total_orders}")
56 print(f"Total Quantity Sold: {total_quantity}")
57 print(f"Average Order Value: ₹{average_sales:,.2f}")
58 print(f"Best Selling Product: {best_product}")
59
60
61 # PRODUCT ANALYSIS
62
63 product_sales = df.groupby('Product')['Total_Sales'].sum().sort_values(ascending=False)
64
65 print("\nSales by Product")
66 print(product_sales)
67
68
69 # REGION ANALYSIS
70
71 region_sales = df.groupby('Region')['Total_Sales'].sum().sort_values(ascending=False)
72
73 print("\nSales by Region")
74 print(region_sales)
75
76
77 # MONTHLY SALES
78
79 monthly_sales = df.groupby('Month')['Total_Sales'].sum()
80
81 month_order = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
82               'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']
83
84 monthly_sales = monthly_sales.reindex(month_order)
85
86 print("\nMonthly Sales")
87 print(monthly_sales)
88
89
90 # CHART 1
91 # Product Sales
92
93 plt.figure(figsize=(8,5))
94 product_sales.plot(kind='bar')
95
96 plt.title("Sales by Product")
97 plt.xlabel("Product")
98 plt.ylabel("Total Sales")
99 plt.xticks(rotation=45)
100
101 plt.tight_layout()
102 plt.savefig("product_sales.png")
103 plt.show()
104

```

main.py > ...

```
105
106 # CHART 2
107 # Monthly Sales Trend
108
109 plt.figure(figsize=(10,5))
110 monthly_sales.plot(kind='line', marker='o')
111
112 plt.title("Monthly Sales Trend")
113 plt.xlabel("Month")
114 plt.ylabel("Total Sales")
115
116 plt.grid(True)
117
118 plt.tight_layout()
119 plt.savefig("monthly_sales.png")
120 plt.show()
121
122
123 # CHART 3
124 # Region Sales
125
126 plt.figure(figsize=(6,6))
127 region_sales.plot(kind='pie', autopct='%1.1f%%')
128
129 plt.title("Regional Sales Distribution")
130 plt.ylabel("")
131
132 plt.tight_layout()
133 plt.savefig("region_sales.png")
134 plt.show()
135
136
```

```

main.py > ...
136
137 # CHART 4
138 # Quantity Distribution
139
140 plt.figure(figsize=(8,5))
141 plt.hist(df['Quantity'], bins=10)
142
143 plt.title("Quantity Distribution")
144 plt.xlabel("Quantity")
145 plt.ylabel("Frequency")
146
147 plt.tight_layout()
148 plt.savefig("quantity_distribution.png")
149 plt.show()
150
151
152 # CHART 5
153 # Price vs Total Sales
154
155 plt.figure(figsize=(8,5))
156 plt.scatter(df['Price'], df['Total_Sales'])
157
158 plt.title("Price vs Total Sales")
159 plt.xlabel("Price")
160 plt.ylabel("Total Sales")
161
162 plt.tight_layout()
163 plt.savefig("price_vs_sales.png")
164 plt.show()
165
166
167 # BUSINESS INSIGHTS
168
169 print("\n----- BUSINESS INSIGHTS ----- \n")
170
171 print(f"1. Total Revenue Generated: ₹{total_sales:,.2f}")
172
173 print(f"2. Highest Selling Product: {product_sales.idxmax()}")
174
175 print(f"3. Highest Revenue Region: {region_sales.idxmax()}")
176 (variable) total_quantity: Any
177 print(f"4. Total Quantity Sold: {total_quantity}")
178
179 print(f"5. Average Order Value: ₹{average_sales:,.2f}")
180
181 print("\nAnalysis Completed Successfully!")

```



## Output:-

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\goura\OneDrive\Desktop\Internship26\E-Commerce_sales_Analysis_Project> & C:\
rive/Desktop/Internship26/E-Commerce_sales_Analysis_Project/main.py
=====
First 5 Rows
=====

```

|   | Date       | Product    | Quantity | Price | Customer_ID | Region | Total_Sales |
|---|------------|------------|----------|-------|-------------|--------|-------------|
| 0 | 2024-01-01 | Phone      | 7        | 37300 | CUST001     | East   | 261100      |
| 1 | 2024-01-02 | Headphones | 4        | 15406 | CUST002     | North  | 61624       |
| 2 | 2024-01-03 | Phone      | 2        | 21746 | CUST003     | West   | 43492       |
| 3 | 2024-01-04 | Headphones | 1        | 30895 | CUST004     | East   | 30895       |
| 4 | 2024-01-05 | Laptop     | 8        | 39835 | CUST005     | North  | 318680      |

```

Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   Date            100 non-null    str
 1   Product         100 non-null    str
 2   Quantity        100 non-null    int64
 3   Price           100 non-null    int64
 4   Customer_ID     100 non-null    str
 5   Region          100 non-null    str
 6   Total_Sales     100 non-null    int64
dtypes: int64(3), str(4)
memory usage: 5.6 KB
None

Missing Values
Date            0
Product         0
Quantity        0
Price           0
Customer_ID     0
Region          0
Total_Sales     0
dtype: int64
```

## Summary Statistics

|       | Date                | Quantity   | Price        | Total_Sales   |
|-------|---------------------|------------|--------------|---------------|
| count | 100                 | 100.000000 | 100.000000   | 100.000000    |
| mean  | 2024-02-19 12:00:00 | 4.780000   | 25808.510000 | 123650.480000 |
| min   | 2024-01-01 00:00:00 | 1.000000   | 1308.000000  | 6540.000000   |
| 25%   | 2024-01-25 18:00:00 | 2.750000   | 14965.250000 | 39517.500000  |
| 50%   | 2024-02-19 12:00:00 | 5.000000   | 24192.000000 | 97955.500000  |
| 75%   | 2024-03-15 06:00:00 | 7.000000   | 38682.250000 | 175792.500000 |
| max   | 2024-04-09 00:00:00 | 9.000000   | 49930.000000 | 373932.000000 |
| std   | NaN                 | 2.588163   | 13917.630242 | 100161.085275 |

===== KPIs =====

Total Sales: ₹12,365,048.00

Total Orders: 100

Total Quantity Sold: 478

Average Order Value: ₹123,650.48

Best Selling Product: Laptop

## Sales by Product

Product

Laptop 3889210

Tablet 2884340

Phone 2859394

Headphones 1384033

Monitor 1348071

Name: Total\_Sales, dtype: int64

## Sales by Region

Region

North 3983635

South 3737852

East 2519639

West 2123922

Name: Total\_Sales, dtype: int64

## Monthly Sales

Month

Jan 4120524.0

Feb 2656050.0

Mar 4485006.0

Apr 1103468.0

May NaN

Jun NaN

Jul NaN

Aug NaN

Sep NaN

Oct NaN

Nov NaN

Dec NaN

Name: Total\_Sales, dtype: float64

----- BUSINESS INSIGHTS -----

1. Total Revenue Generated: ₹12,365,048.00
2. Highest Selling Product: Laptop
3. Highest Revenue Region: North
4. Total Quantity Sold: 478
5. Average Order Value: ₹123,650.48

Analysis Completed Successfully!

PS C:\Users\goura\OneDrive\Desktop\Internship26\E-Commerce\_sales\_Analysis\_Project>

# Jupyter notebook outputs:-

```
[4]: import pandas as pd
import matplotlib.pyplot as plt
```

```
[5]: #Load dataset
df = pd.read_csv("data/sales_data.csv")

print("="*50)
print("First 5 Rows")
print("="*50)
print(df.head())
```

```
=====
First 5 Rows
=====
```

|   | Date       | Product    | Quantity | Price | Customer_ID | Region | Total_Sales |
|---|------------|------------|----------|-------|-------------|--------|-------------|
| 0 | 2024-01-01 | Phone      | 7        | 37300 | CUST001     | East   | 261100      |
| 1 | 2024-01-02 | Headphones | 4        | 15406 | CUST002     | North  | 61624       |
| 2 | 2024-01-03 | Phone      | 2        | 21746 | CUST003     | West   | 43492       |
| 3 | 2024-01-04 | Headphones | 1        | 30895 | CUST004     | East   | 30895       |
| 4 | 2024-01-05 | Laptop     | 8        | 39835 | CUST005     | North  | 318680      |

```
[6]: print("\nDataset Information")
print(df.info())
```

```
Dataset Information
<class 'pandas.DataFrame'>
RangeIndex: 100 entries, 0 to 99
Data columns (total 7 columns):
#   Column          Non-Null Count  Dtype
---  ---
0   Date             100 non-null   str
1   Product          100 non-null   str
2   Quantity         100 non-null   int64
3   Price            100 non-null   int64
4   Customer_ID      100 non-null   str
5   Region           100 non-null   str
6   Total_Sales      100 non-null   int64
dtypes: int64(3), str(4)
memory usage: 5.6 KB
None
```

```
[7]: #Checking missing values
```

```
print("\nMissing Values")
print(df.isnull().sum())
```

```
Missing Values
Date          0
Product       0
Quantity      0
Price         0
Customer_ID   0
Region        0
Total_Sales   0
dtype: int64
```

```
[22]: # DATA CLEANING

# Remove duplicate rows
df.drop_duplicates(inplace=True)

[9]: # Convert Date column to datetime
df['Date'] = pd.to_datetime(df['Date'])

[10]: # Create Month column
df['Month'] = df['Date'].dt.strftime('%b')

[11]: print("\nSummary Statistics")
print(df.describe())
```

```
Summary Statistics
```

|       | Date                | Quantity   | Price        | Total_Sales   |
|-------|---------------------|------------|--------------|---------------|
| count | 100                 | 100.000000 | 100.000000   | 100.000000    |
| mean  | 2024-02-19 12:00:00 | 4.780000   | 25808.510000 | 123650.480000 |
| min   | 2024-01-01 00:00:00 | 1.000000   | 1308.000000  | 6540.000000   |
| 25%   | 2024-01-25 18:00:00 | 2.750000   | 14965.250000 | 39517.500000  |
| 50%   | 2024-02-19 12:00:00 | 5.000000   | 24192.000000 | 97955.500000  |
| 75%   | 2024-03-15 06:00:00 | 7.000000   | 38682.250000 | 175792.500000 |
| max   | 2024-04-09 00:00:00 | 9.000000   | 49930.000000 | 373932.000000 |
| std   | NaN                 | 2.588163   | 13917.630242 | 100161.085275 |

```
[23]: # KPI ANALYSIS

total_sales = df['Total_Sales'].sum()
total_orders = len(df)
total_quantity = df['Quantity'].sum()
average_sales = df['Total_Sales'].mean()

#Best Selling Product
best_product = (
    df.groupby("Product")["Quantity"]
    .sum()
    .idxmax()
)

print("\n===== KPIs =====")
print(f"Total Sales: ₹{total_sales:,.2f}")
print(f"Total Orders: {total_orders}")
print(f"Total Quantity Sold: {total_quantity}")
print(f"Average Order Value: ₹{average_sales:,.2f}")
print(f"Best Selling Product: {best_product}")
```

```
===== KPIs =====
Total Sales: ₹12,365,048.00
Total Orders: 100
Total Quantity Sold: 478
Average Order Value: ₹123,650.48
Best Selling Product: Laptop
```

[13]: # PRODUCT ANALYSIS

```
product_sales = df.groupby('Product')['Total_Sales'].sum().sort_values(ascending=False)

print("\nSales by Product")
print(product_sales)
```

```
Sales by Product
Product
Laptop      3889210
Tablet      2884340
Phone       2859394
Headphones  1384033
Monitor     1348071
Name: Total_Sales, dtype: int64
```

[14]: # REGION ANALYSIS

```
region_sales = df.groupby('Region')['Total_Sales'].sum().sort_values(ascending=False)

print("\nSales by Region")
print(region_sales)
```

```
Sales by Region
Region
North   3983635
South   3737852
East    2519639
West    2123922
Name: Total_Sales, dtype: int64
```

[15]: # MONTHLY SALES

```
monthly_sales = df.groupby('Month')['Total_Sales'].sum()

month_order = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun',
               'Jul', 'Aug', 'Sep', 'Oct', 'Nov', 'Dec']

monthly_sales = monthly_sales.reindex(month_order)

print("\nMonthly Sales")
print(monthly_sales)
```

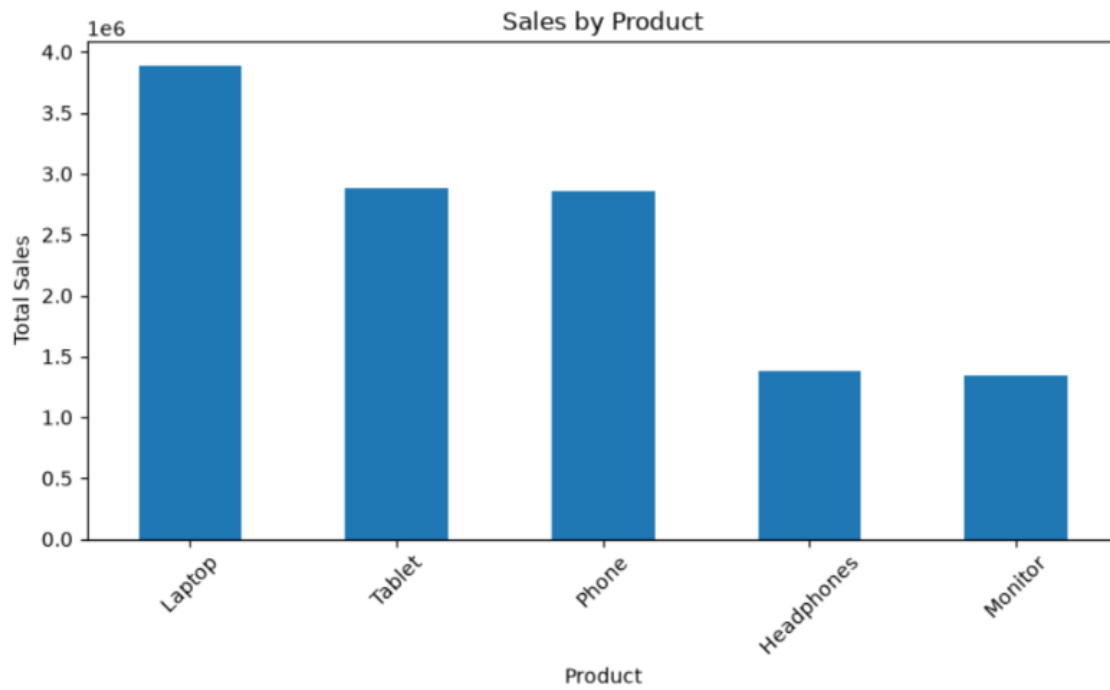
```
Monthly Sales
Month
Jan      4120524.0
Feb      2656050.0
Mar      4485006.0
Apr      1103468.0
May              NaN
Jun              NaN
Jul              NaN
Aug              NaN
Sep              NaN
Oct              NaN
Nov              NaN
Dec              NaN
Name: Total_Sales, dtype: float64
```

```
[16]: # CHART 1
# Product Sales

plt.figure(figsize=(8,5))
product_sales.plot(kind='bar')

plt.title("Sales by Product")
plt.xlabel("Product")
plt.ylabel("Total Sales")
plt.xticks(rotation=45)

plt.tight_layout()
plt.savefig("product_sales.png")
plt.show()
```



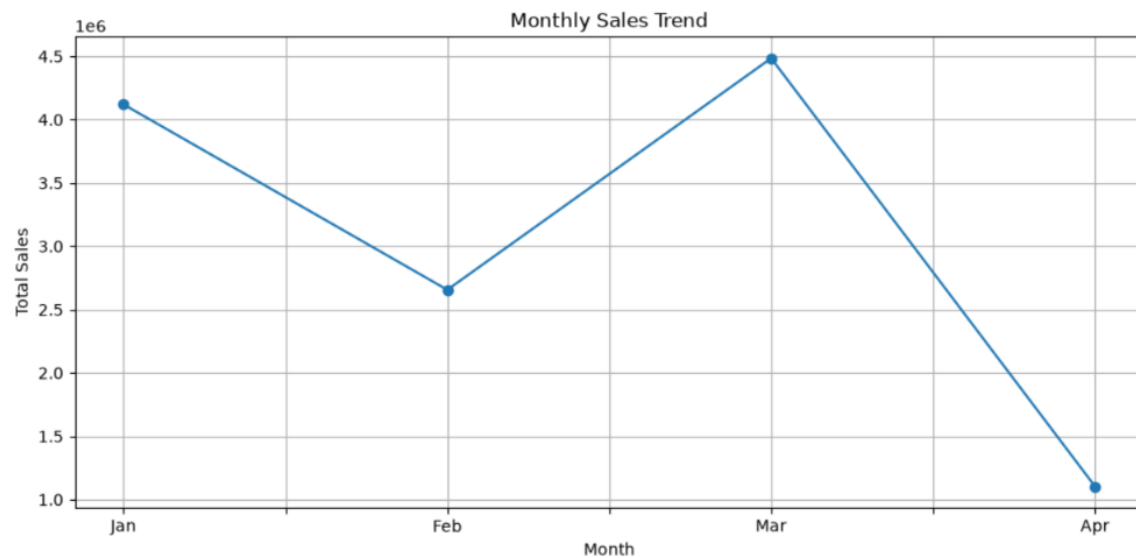
```
[17]: # CHART 2
# Monthly Sales Trend

plt.figure(figsize=(10,5))
monthly_sales.plot(kind='line', marker='o')

plt.title("Monthly Sales Trend")
plt.xlabel("Month")
plt.ylabel("Total Sales")

plt.grid(True)

plt.tight_layout()
plt.savefig("monthly_sales.png")
plt.show()
```



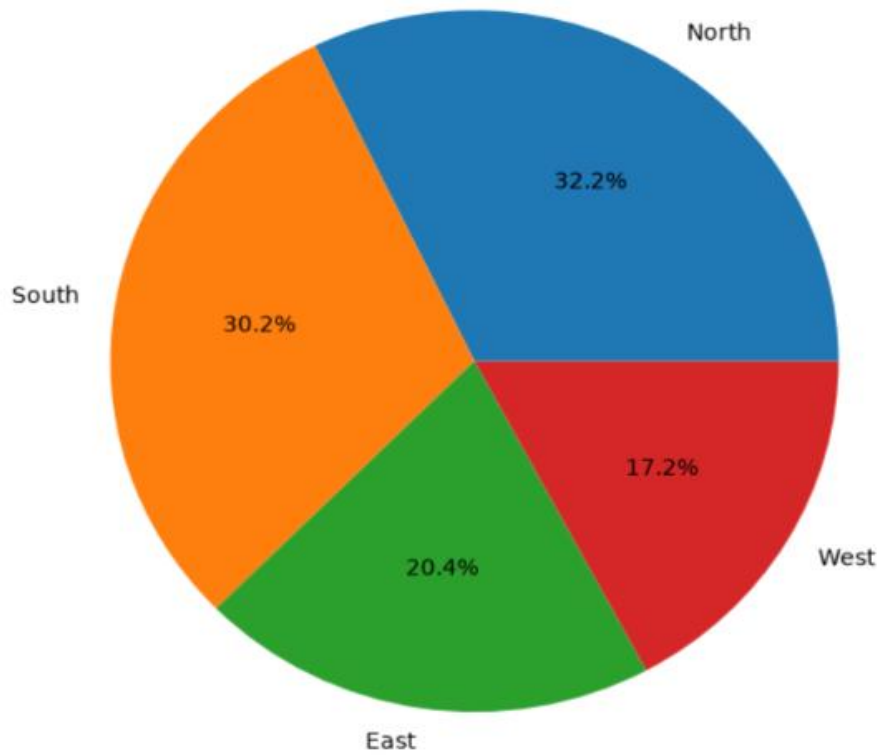
```
[18]: # CHART 3
# Region Sales

plt.figure(figsize=(6,6))
region_sales.plot(kind='pie', autopct='%1.1f%%')

plt.title("Regional Sales Distribution")
plt.ylabel("")

plt.tight_layout()
plt.savefig("region_sales.png")
plt.show()
```

Regional Sales Distribution



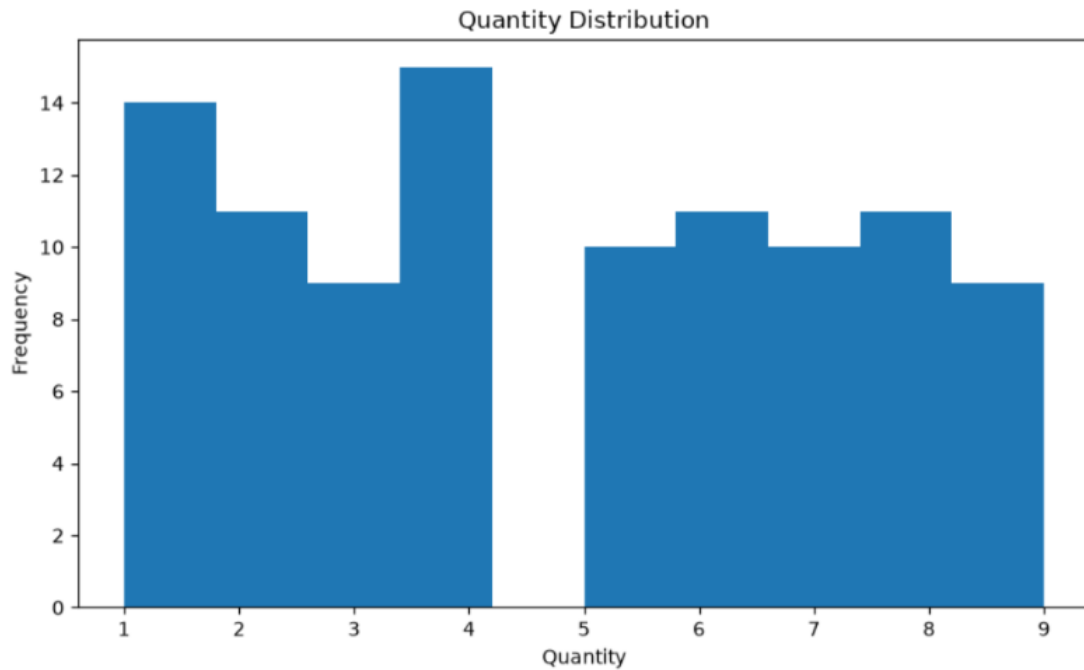
```
19]: # CHART 4
      # Quantity Distribution

      plt.figure(figsize=(8,5))
      plt.hist(df['Quantity'], bins=10)

      plt.title("Quantity Distribution")
      plt.xlabel("Quantity")
      plt.ylabel("Frequency")

      plt.tight_layout()
      plt.savefig("quantity_distribution.png")
      plt.show()
```





```
[20]: # CHART 5
# Price vs Total Sales

plt.figure(figsize=(8,5))
plt.scatter(df['Price'], df['Total_Sales'])

plt.title("Price vs Total Sales")
plt.xlabel("Price")
plt.ylabel("Total Sales")

plt.tight_layout()
plt.savefig("price_vs_sales.png")
plt.show()
```



```
[21]: # BUSINESS INSIGHTS

print("\n----- BUSINESS INSIGHTS ----- \n")

print(f"1. Total Revenue Generated: ₹{total_sales:,.2f}")

print(f"2. Highest Selling Product: {product_sales.idxmax()}")

print(f"3. Highest Revenue Region: {region_sales.idxmax()}")

print(f"4. Total Quantity Sold: {total_quantity}")

print(f"5. Average Order Value: ₹{average_sales:,.2f}")

print("\nAnalysis Completed Successfully!")

----- BUSINESS INSIGHTS -----

1. Total Revenue Generated: ₹12,365,048.00
2. Highest Selling Product: Laptop
3. Highest Revenue Region: North
4. Total Quantity Sold: 478
5. Average Order Value: ₹123,650.48

Analysis Completed Successfully!
```

## Learning outcomes

Through this project, I gained practical experience in:

- Data Cleaning
- Exploratory Data Analysis (EDA)
- Data Visualization
- Business Analytics
- Python Programming
- Data Interpretation
- Report Writing

# Conclusion

This project successfully analyzed an e-commerce sales dataset using Python. The analysis included data cleaning, KPI calculation, exploratory data analysis, and visualization using matplotlib. The generated insights help identify top-performing products and regions, providing valuable information for business decision-making.

The project demonstrates practical skills in Python programming, data analysis, and visualization, making it suitable for real-world business analytics tasks and internship evaluation.

# References

- Python official documentation
- Google
- Chatgpt
- Youtube
- Project guidance provided in tasks.